

Introduction to Module 6: Hashing

Design and Analysis of Algorithms

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Main Topics

- Data structures:
 - Hash tables and their uses as fast sets and maps
 - “Universal” hash functions for achieving strong performance bounds for hash tables
 - Distributed hash tables and low-memory “sketch” structures
- Very wide range of additional applications:
 - String matching, security, networking, streaming, large-scale problem decomposition, random number generation, data integrity.
- Coding discussion: Hash tables in action
- Optimal enrichment lecture: Locality-sensitive hashing

Algorithm Design Assignment

- Design, Describe, and Analyze algorithms.
- Practice using and combining tools and basic algorithms.
- Algorithms should be described concisely in English (possibly supported by pseudocode when helpful), making sure key details are present. Figures welcome.
- Solutions should be typeset.
- Collaboration in small groups encouraged, but your final write-up should be your own.